set of axioms for reasoning with degrees of beliefs: no contradictions, correspondence with ordinary logic (for example, if belief in A goes up, then belief in $\neg A$ must go down), and so on. The only controversial axiom is that degrees of belief must be numbers, or at least act like numbers in that they must be transitive (if belief in A is greater than belief in B, which is greater than belief in C, then belief in A must be greater than C) and comparable (the belief in A must be one of equal to, greater than, or less than belief in B). It can then be proved that probability is the only approach that satisfies these axioms.

The world being the way it is, however, practical demonstrations sometimes speak louder than proofs. The success of reasoning systems based on probability theory has been much more effective than philosophical arguments in making converts. We now look at how the axioms can be deployed to make inferences.

12.3 Inference Using Full Joint Distributions

Probabilistic inference
Query

In this section we describe a simple method for **probabilistic inference**—that is, the computation of posterior probabilities for **query** propositions given observed evidence. We use the full joint distribution as the "knowledge base" from which answers to all questions may be derived. Along the way we also introduce several useful techniques for manipulating equations involving probabilities.

We begin with a simple example: a domain consisting of just the three Boolean variables *Toothache*, *Cavity*, and *Catch* (the dentist's nasty steel probe catches in my tooth). The full joint distribution is a $2 \times 2 \times 2$ table as shown in Figure 12.3.

	toothache		$\neg$*toothache*	
	catch	$\neg$*catch*	*catch*	$\neg$*catch*
cavity	0.108	0.012	0.072	0.008
$\neg$*cavity*	0.016	0.064	0.144	0.576

Figure 12.3 A full joint distribution for the *Toothache*, *Cavity*, *Catch* world.

Notice that the probabilities in the joint distribution sum to 1, as required by the axioms of probability. Notice also that Equation (12.2) gives us a direct way to calculate the probability of any proposition, simple or complex: simply identify those possible worlds in which the proposition is true and add up their probabilities. For example, there are six possible worlds in which $cavity \vee toothache$ holds:

$$P(cavity \vee toothache) = 0.108 + 0.012 + 0.072 + 0.008 + 0.016 + 0.064 = 0.28\,.$$

One particularly common task is to extract the distribution over some subset of variables or a single variable. For example, adding the entries in the first row gives the unconditional or **marginal probability**[6] of *cavity*:

Marginal probability

$$P(cavity) = 0.108 + 0.012 + 0.072 + 0.008 = 0.2\,.$$

[6] So called because of a common practice among actuaries of writing the sums of observed frequencies in the margins of insurance tables.

Marginalization

This process is called **marginalization**, or **summing out**—because we sum up the probabilities for each possible value of the other variables, thereby taking them out of the equation. We can write the following general marginalization rule for any sets of variables $\mathbf{Y}$ and $\mathbf{Z}$:

$$\mathbf{P}(\mathbf{Y}) = \sum_{\mathbf{z}} \mathbf{P}(\mathbf{Y}, \mathbf{Z}=\mathbf{z}), \tag{12.7}$$

where $\sum_{\mathbf{z}}$ sums over all the possible combinations of values of the set of variables $\mathbf{Z}$. As usual we can abbreviate $\mathbf{P}(\mathbf{Y}, \mathbf{Z}=\mathbf{z})$ in this equation by $\mathbf{P}(\mathbf{Y}, \mathbf{z})$. For the *Cavity* example, Equation (12.7) corresponds to the following equation:

$$\begin{aligned}\mathbf{P}(Cavity) &= \mathbf{P}(Cavity, toothache, catch) + \mathbf{P}(Cavity, toothache, \neg catch) \\ &+ \mathbf{P}(Cavity, \neg toothache, catch) + \mathbf{P}(Cavity, \neg toothache, \neg catch) \\ &= \langle 0.108, 0.016\rangle + \langle 0.012, 0.064\rangle + \langle 0.072, 0.144\rangle + \langle 0.008, 0.576\rangle \\ &= \langle 0.2, 0.8\rangle .\end{aligned}$$

Conditioning

Using the product rule (Equation (12.4)), we can replace $\mathbf{P}(\mathbf{Y}, \mathbf{z})$ in Equation (12.7) by $\mathbf{P}(\mathbf{Y} \mid \mathbf{z})P(\mathbf{z})$, obtaining a rule called **conditioning**:

$$\mathbf{P}(\mathbf{Y}) = \sum_{\mathbf{z}} \mathbf{P}(\mathbf{Y} \mid \mathbf{z})P(\mathbf{z}). \tag{12.8}$$

Marginalization and conditioning turn out to be useful rules for all kinds of derivations involving probability expressions.

In most cases, we are interested in computing *conditional* probabilities of some variables, given evidence about others. Conditional probabilities can be found by first using Equation (12.3) to obtain an expression in terms of unconditional probabilities and then evaluating the expression from the full joint distribution. For example, we can compute the probability of a cavity, given evidence of a toothache, as follows:

$$\begin{aligned}P(cavity \mid toothache) &= \frac{P(cavity \wedge toothache)}{P(toothache)} \\ &= \frac{0.108 + 0.012}{0.108 + 0.012 + 0.016 + 0.064} = 0.6 .\end{aligned}$$

Just to check, we can also compute the probability that there is no cavity, given a toothache:

$$\begin{aligned}P(\neg cavity \mid toothache) &= \frac{P(\neg cavity \wedge toothache)}{P(toothache)} \\ &= \frac{0.016 + 0.064}{0.108 + 0.012 + 0.016 + 0.064} = 0.4 .\end{aligned}$$

The two values sum to 1.0, as they should. Notice that the term $P(toothache)$ is in the denominator for both of these calculations. If the variable *Cavity* had more than two values, it would be in the denominator for all of them. In fact, it can be viewed as a **normalization** constant for the distribution $\mathbf{P}(Cavity \mid toothache)$, ensuring that it adds up to 1. Throughout the chapters dealing with probability, we use α to denote such constants. With this notation, we can write the two preceding equations in one:

$$\begin{aligned}&\mathbf{P}(Cavity \mid toothache) = \alpha \mathbf{P}(Cavity, toothache) \\ &= \alpha [\mathbf{P}(Cavity, toothache, catch) + \mathbf{P}(Cavity, toothache, \neg catch)] \\ &= \alpha [\langle 0.108, 0.016\rangle + \langle 0.012, 0.064\rangle] = \alpha \langle 0.12, 0.08\rangle = \langle 0.6, 0.4\rangle .\end{aligned}$$